MINIMUM-VARIANCE PHYLOGENETIC TREES ARE NOT UNIQUE UP TO DEPTH-EQUIVALENCE

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Abstract. Coronado, Mir, Rosselló and Rotger studied the variance $V(T)$ of the depths of the leaves of a rooted bifurcating tree T, the balance index originally proposed by Sackin. They gave an algorithm for the minimum of V over the bifurcating trees with n leaves, ran it for every $n \leq 2^{20}$, and conjectured that for every n the minimizing trees are unique up to depth-equivalence, that is, they all have the same multiset of leaf depths. We show that the conjecture is false. For $n = 67{,}256{,}579$ there are two bifurcating trees with different multisets of leaf depths that both attain the minimum, $n^2V = 330{,}431{,}494{,}786$. Using the structure theorem of Coronado et al., an exact search shows that this is the smallest such n, and that up to 2^{28} there are exactly six values of n with two minimizing profiles. The proof of global minimality is self-contained. Cutting an arbitrary tree at a suitable level reduces the problem to a relaxed integer problem whose optimal solutions have a rigid normal form, and a certificate of 16,800 exact integer evaluations completes the argument. The disproof, including the construction of the two trees and their global minimality among all bifurcating trees with n leaves, is formalized and checked in Lean 4.

1. Introduction

Balance indices quantify how symmetric the shape of a rooted phylogenetic tree is, and they are used to test models of evolution against the shapes of observed trees. As recalled in [1], Sackin originally proposed to measure the imbalance of a tree by the "variation" of the depths of its leaves, and Kirkpatrick and Slatkin implemented this proposal as their variance. For a rooted bifurcating tree T with n leaves of depths $d_1, \ldots, d_n$, this variance is

$$V(T) = \frac{1}{n}\sum_{i=1}^{n} d_i^2 - \Big(\frac{1}{n}\sum_{i=1}^{n} d_i\Big)^2.$$

Coronado, Mir, Rosselló and Rotger [1] studied V as a balance index. They showed that its maximum over the bifurcating trees with n leaves is attained exactly at the combs. Its minimum is attained at the maximally balanced trees for $n \leq 183$, but not for almost all larger n. They proved that every tree with minimum V belongs to an explicit family of tree types (Theorem 2.1 below), obtained from it an algorithm computing the minimum, and ran the algorithm for every $n \leq 2^{20}$. In the Discussion of [1] they list regularities observed in these computations; the first one is the following.

Conjecture 1.1 ([1, Discussion])**.** *For every n, the tree in $\mathcal{BT}_n^*$ with minimum V value is unique up to depth-equivalence.*

Here $\mathcal{BT}_n^*$ is the set of rooted bifurcating tree shapes with n leaves, and two trees are depth-equivalent when they have the same multiset of leaf depths. The characterization and the number of the trees of minimum V are listed as open problems in the survey of tree balance indices by Fischer, Herbst, Kersting, Kühn and Wicke [2].

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Theorem 1.2. *Let $n = 67{,}256{,}579$. The minimum of $n^2V(T)$ over all rooted bifurcating trees T with n leaves is 330,431,494,786. It is attained by two trees T_A and T_B whose multisets of leaf depths are*

$$\Delta(T_A) = \{2,3,4,5,6,7,8,9,11,12,14,15,16,17,18,19\} \uplus \{26^{13},\, 27^{67,256,550}\},$$
$$\Delta(T_B) = \{2,3,4,5,6,7,8,9,11,12,13\} \uplus \{27^{67,256,072},\, 28^{496}\},$$

where d^c denotes c leaves of depth d. In particular, Conjecture 1.1 is false.

The two trees are not depth-equivalent: only T_B has leaves of depth 28. Since $n = 2^{26} + 147{,}715$, the counterexample lies far beyond the range $n \le 2^{20}$ covered by the computations of [1]. Using the structure theorem of [1], we also determined by an exact computation all $n \le 2^{28}$ for which the minimizer is not unique up to depth-equivalence (Section 5). There are exactly six, namely

$$67{,}256{,}579, \quad 140{,}525{,}827, \quad 143{,}671{,}555, \quad 151{,}535{,}875, \quad 168{,}050{,}947, \quad 201{,}474{,}307,$$

and for each of them there are exactly two minimizing profiles. In particular, $n = 67{,}256{,}579$ is the smallest counterexample to Conjecture 1.1, and the computations of [1] for $n \le 2^{20}$ are confirmed.

The proof of minimality in Section 4 does not use the structure theorem of [1]. It bounds the centred second moment of a hypothetical better tree, cuts the tree at a suitable level, and relaxes the problem to an integer problem about "coins" whose optimal solutions have a rigid normal form. A finite certificate then shows that no tree beats T_A and T_B. Section 6 describes the Lean formalization, which checks the whole argument.

2. Notation

A rooted bifurcating tree is a rooted tree in which every internal vertex has exactly two children; the tree with one vertex is allowed. The depth of a vertex is the number of edges on the path from the root to it. For a tree T with n leaves of depths $d_1, \dots, d_n$, let $\Delta(T)$ be the multiset $\{d_1, \dots, d_n\}$ and put

$$S(T) = \sum_i d_i, \qquad Q(T) = \sum_i d_i^2, \qquad D(T) = nQ(T) - S(T)^2 = n^2V(T).$$

For fixed n, minimizing V is the same as minimizing the integer D. For an integer k we use the centred moments

$$E_k(T) = \sum_i (k - d_i) = kn - S(T), \qquad R_k(T) = \sum_i (k - d_i)^2.$$

Since the variance does not change under translation,

$$D(T) = nR_k(T) - E_k(T)^2 \qquad \text{for every integer } k. \tag{1}$$

A multiset $\{d_1, \dots, d_n\}$ of non-negative integers is the multiset of leaf depths of a rooted bifurcating tree if and only if $\sum_i 2^{-d_i} = 1$ (Kraft's equality). Necessity follows by induction on the tree. For sufficiency, let h be the largest depth; multiplying the equality by 2^h shows that the number of leaves of depth h is even, so they can be paired into cherries whose parents have depth $h - 1$, and induction on $\sum_i 2^{d_i}$ finishes the construction.

The structure theorem of [1] reads as follows. Following [1, Definition 3], for $j \ge 0$ and $2 \le l_1 < \cdots < l_j \le \delta - 2$, a tree of depth δ with n leaves is *of type* $T_{n;l_1,\dots,l_j}$ if it has exactly one leaf of depth $\delta - l_i$ for each $i = 1, \dots, j$ and all its other leaves have depth δ or $\delta - 1$.

Theorem 2.1 ([1, Theorem 2])**.** *If $T \in \mathcal{BT}_n^*$ has the minimum value of V, then it is of some type $T_{n;l_1,\dots,l_j}$ with $5 \le l_1 < \cdots < l_j \le \delta(T) - 2$.*

We use Theorem 2.1 only for the independent computations of Section 5.

3. The two trees

From now on $n = 67{,}256{,}579$. Put

$$A = \{8, 9, 10, 11, 12, 13, 15, 16, 18, 19, 20, 21, 22, 23, 24, 25\},$$
$$B = \{14, 15, 16, 18, 19, 20, 21, 22, 23, 24, 25\}.$$

The profile $\Delta(T_A)$ consists of one leaf of depth $27 - t$ for each $t \in A$, 13 leaves of depth 26 and 67,256,550 leaves of depth 27. The profile $\Delta(T_B)$ consists of one leaf of depth $27 - t$ for each $t \in B$, 67,256,072 leaves of depth 27 and 496 leaves of depth 28. Both have n elements, and both satisfy Kraft's equality:

$$\sum_{t\in A} 2^t + 13 \cdot 2 + 67{,}256{,}550 = 2^{27}, \qquad \sum_{t\in B} 2^t + 67{,}256{,}072 + \frac{496}{2} = 2^{27}.$$

Concretely, T_A can be built from a path $v_0, v_1, \ldots, v_{22}$ starting at the root v_0. For $t = 26, 25, \ldots, 5$, the vertex v_{26-t} receives as its second child, at depth $27 - t$, a single leaf if $t \in A$ and a perfect binary tree of height t otherwise. Finally, v_{22} is the root of a tree with 13 leaves of depth 26 and 6 leaves of depth 27. The tree T_B is built in the same way from a path $v_0, \ldots, v_{18}$ with $t = 26, \ldots, 9$ and the set B, and v_{18} is the root of a tree with 264 leaves of depth 27 and 496 of depth 28. In the notation of [1], T_A is of type $T_{n;8,9,10,11,12,13,15,16,18,19,20,21,22,23,24,25}$ (depth 27) and T_B is of type $T_{n;15,16,17,19,20,21,22,23,24,25,26}$ (depth 28).

With respect to the level $k = 27$,

$$E_{27}(T_A) = \sum_{t\in A} t + 13 = 279, \qquad R_{27}(T_A) = \sum_{t\in A} t^2 + 13 = 4913,$$
$$E_{27}(T_B) = \sum_{t\in B} t - 496 = -279, \qquad R_{27}(T_B) = \sum_{t\in B} t^2 + 496 = 4913.$$

By (1) both trees have

$$D = 4913\, n - 279^2 = 330{,}431{,}494{,}786.$$

Thus the two trees tie because their centred first moments are opposite and their centred second moments agree.

4. Global minimality

Put $C = 4913$ and $D^* = Cn - 279^2$. We prove that every bifurcating tree T with n leaves satisfies $D(T) \geq D^*$; together with Section 3 this proves Theorem 1.2. So let T have n leaves and suppose $D(T) \leq D^*$.

4.1. **A bound on the centred second moment.**

Lemma 4.1. *Let k be a non-negative integer nearest to $S(T)/n$, and write $E = E_k(T)$, $R = R_k(T)$. Then $R \leq C$ and $27 \leq k \leq 96$.*

Proof. By the choice of k we have $|E| \leq n/2$. Since $|z| \leq z^2$ for every integer z, also $|E| \leq \sum_i |k - d_i| \leq R$. Hence $E^2 \leq Rn/2$, so $D(T) = nR - E^2 \geq nR/2$ and $R \leq 2D^*/n < 2C$. Suppose $R \geq C + 1$. Using $E^2 \leq R^2$ and the fact that $u \mapsto nu - u^2$ is increasing for $u < n/2$, we get

$$D(T) \geq nR - R^2 \geq (C + 1)n - (C + 1)^2 > Cn - 279^2,$$

where the last inequality holds because $n > (C + 1)^2 - 279^2 = 24{,}069{,}555$. This contradicts $D(T) \leq D^*$, so $R \leq C$.

For the range of k we use the cut at level k described next. The relations (3) and (4) below give $2x \le Q_{\mathrm{deep}} \le R \le C$, so $x \le 2456$, and (2) gives $n \le 2^k + x$. Since $n - 2456 > 2^{26}$, this forces $k \ge 27$. On the other hand, a bifurcating tree whose shallowest leaf has depth $d_{\min}$ has at least $2^{d_{\min}}$ leaves, so T has a leaf of depth at most 26. That leaf contributes $(k-26)^2$ or more to R, so $(k-26)^2 \le C < 71^2$ and $k \le 96$. □

4.2. **Cutting at a level.** Fix an integer $k \ge 0$. Let M be the multiset of the gaps $t = k - d \ge 1$ of the leaves of T of depth $d < k$, the *shallow* leaves. The vertices of T at depth exactly k are the roots of disjoint subtrees $U_1, \dots, U_p$, which contain all the other leaves. Let m_u be the number of leaves of U_u, and put

$$x = \sum_u (m_u - 1), \qquad S_{\mathrm{deep}} = \sum_u S(U_u), \qquad Q_{\mathrm{deep}} = \sum_u Q(U_u),$$

with depths in U_u measured from its root. In the perfect binary tree of height k, a shallow leaf with gap t occupies 2^t positions at level k and each root of a subtree U_u occupies one, so $\sum_{t\in M} 2^t + p = 2^k$. Since $n = |M| + \sum_u m_u = |M| + p + x$, this becomes

$$\sum_{t\in M} \left(2^t - 1\right) = 2^k - n + x. \tag{2}$$

For a bifurcating tree U with m leaves we have $2(m-1) \le S(U) \le Q(U)$: the first inequality holds because $S(U)$ is the sum, over the $m-1$ internal vertices, of their numbers of descendant leaves, each at least 2; the second because $d \le d^2$. Summing over the subtrees,

$$2x \le S_{\mathrm{deep}} \le Q_{\mathrm{deep}}, \tag{3}$$

and directly from the definitions,

$$E_k(T) = \sum_{t\in M} t - S_{\mathrm{deep}}, \qquad R_k(T) = \sum_{t\in M} t^2 + Q_{\mathrm{deep}}. \tag{4}$$

Consequently the *coin cost*

$$c(M, x) = \sum_{t\in M} t^2 + 2x$$

satisfies $c(M,x) \le R_k(T)$, and if equality holds then $S_{\mathrm{deep}} = Q_{\mathrm{deep}} = 2x$ and

$$E_k(T) = \sum_{t\in M} t - 2x. \tag{5}$$

4.3. **The relaxed coin problem.** Fix an integer $r \ge 0$ and consider all pairs (M, x) of a finite multiset M of positive integers and an integer $x \ge 0$ satisfying the coin equation

$$\sum_{t\in M} \left(2^t - 1\right) = r + x. \tag{6}$$

We call such a pair feasible and minimize $c(M,x)$. Since costs are non-negative integers, a feasible pair of minimum cost exists whenever a feasible pair exists.

Lemma 4.2. *Every feasible pair of minimum cost has the following form: M consists of a copies of 1 together with a set L of distinct integers ≥ 5, and $a = 0$ or $x = 0$.*

Proof. Each of the following replacements preserves (6) and strictly lowers the cost, so none of them applies to a pair of minimum cost. (i) Replace a gap $t \in \{2,3,4\}$ by $2^t - 1$ gaps equal to 1; the cost changes by $(2^t - 1) - t^2 \in \{-1, -2, -1\}$. (ii) Replace two gaps $t \ge 5$ by one gap $t+1$ and increase x by one; since $(2^{t+1} - 1) - 2(2^t - 1) = 1$, (6) is preserved, and the cost changes by $(t+1)^2 + 2 - 2t^2 = -(t-3)(t+1) < 0$. (iii) If $x > 0$ and $1 \in M$, remove one gap 1 and decrease x by one; the cost decreases by 3. □

For a pair in this normal form let $j = |L|$ and let $b = \sum_{t\in L} 2^{t-5}$, so that L is read off from the binary digits of b, and

$$\sum_{t\in L}\bigl(2^t - 1\bigr) = 32b - j =: A(b).$$

The coin equation determines a and x: if $A(b) \le r$ then $x = 0$ and $a = r - A(b)$, and if $A(b) > r$ then $a = 0$ and $x = A(b) - r$. Hence the cost and the right-hand side of (5) are

$$F(r,b) = \sum_{t\in L} t^2 + \max\bigl(r - A(b), 0\bigr) + 2\max\bigl(A(b) - r, 0\bigr),$$
$$G(r,b) = \sum_{t\in L} t + \max\bigl(r - A(b), 0\bigr) - 2\max\bigl(A(b) - r, 0\bigr).$$

If $F(r,b) \le C$, then $a \le C$, $x \le \lfloor C/2 \rfloor = 2456$ and $25j \le C$, so $j \le 196$. Since $32b = r + x - a + j$, it follows that $r - C \le 32b \le r + 2652$, and therefore

$$b = \left\lfloor \frac{r - C}{32} \right\rfloor + i \qquad \text{for some } 0 \le i < 240. \tag{7}$$

4.4. **The certificate.** For $27 \le k \le 96$ and $0 \le i < 240$ put $r = 2^k - n$ (so $r > C$) and let b be given by (7). Evaluating F and G in exact integer arithmetic at these $70 \cdot 240 = 16{,}800$ points shows:

$$F(r,b) \le C \implies k = 27, \quad F(r,b) = C, \quad G(r,b)^2 = 279^2. \tag{8}$$

Exactly two points satisfy the premise, both with $k = 27$: $i = 154$, $b = 2{,}092{,}536$, $L = A$, $G = 279$; and $i = 162$, $b = 2{,}092{,}544$, $L = B$, $G = -279$. They correspond to the trees T_A and T_B.

4.5. **Conclusion of the proof.** Let k be as in Lemma 4.1, and let (M, x) be obtained by cutting T at level k. By (2), (M, x) is feasible for $r = 2^k - n$, and $c(M, x) \le R_k(T) \le C$. Take a feasible pair of minimum cost for this r. It has the normal form of Lemma 4.2, its cost is at most C, and its b satisfies (7); by (8) its cost equals C. Hence the minimum cost is C, so $c(M, x) = C$, and (M, x) itself has minimum cost. Applying Lemma 4.2 and (8) to (M, x) shows that $\sum_{t\in M} t - 2x = G(r,b) = \pm 279$. Since $c(M, x) = C \ge R_k(T) \ge c(M, x)$, equality holds, and by (5) $E_k(T) = \pm 279$. By (1), $D(T) = nC - 279^2 = D^*$. Thus no tree has $D < D^*$, which proves Theorem 1.2.

5. Computations

All computations below use exact integer arithmetic; the scripts accompany the source files.

The certificate. The script `verify_certificate.py` evaluates the 16,800 entries of (8), prints the two successful entries, and checks the two profiles (Kraft's equality, the number of leaves, and D). A re-check of the certificate over the wider window $-300 \le i < 1000$ finds no further solutions; for $k \ge 28$ the smallest value of F is 5640.

The family of Theorem 2.1. Two independent programs enumerate all types $T_{n;l_1,\dots,l_j}$ with $5 \le l_1 < \cdots < l_j \le \delta - 2$ for $\delta \in \{26, 27, 28\}$. Both find 2,101,768 realizable types for $n = 67{,}256{,}579$, with minimum $D = 330{,}431{,}494{,}786$ attained exactly by the types of T_A and T_B. This confirms Theorem 1.2 by a route that uses Theorem 2.1 instead of Section 4.

Small n. An exhaustive enumeration of all leaf-depth profiles for $n \le 24$ agrees with the minimum over the family of Theorem 2.1, with a unique minimizing profile in each case.

All $n \le 2^{28}$. For each n we computed the minimum of D over the family of Theorem 2.1 (types with $\delta \in \{m, m+1, m+2\}$, where $m = \lfloor \log_2 n \rfloor$; indeed $2^\delta \ge n$, and Kraft's equality gives $2^\delta = p_0 + 2p_1 + \sum_i 2^{l_i} < 2n + 2^{\delta-1}$, where p_0 and p_1 are the numbers of leaves of depths δ and

$\delta-1$) and the number of minimizing types. For $5 \le n \le 2^{20}$ we enumerated the whole family (program `minvar_scan.c`); the minimizer is unique for every such n, in agreement with [1]. For larger n we used a pruned enumeration (`minvar_fast.c`) based on the following observation. Let c be an integer nearest to the mean depth of a candidate; then $c \in \{\delta-1, \delta\}$, because the mean depth is $\delta-(p_1+l_1+\cdots+l_j)/n$ with $l_1+\cdots+l_j < m^2 < n/2$, and $D \ge nR_c/2$ as in the proof of Lemma 4.1. If $c=\delta$, then R_c is at least the number p_1 of leaves of depth $\delta-1$, and if $c=\delta-1$, then R_c is at least the number p_0 of leaves of depth δ. So every candidate with $D \le D_{\mathrm{best}}$ has $p_1 \le 2D_{\mathrm{best}}/n$ or $p_0 \le 2D_{\mathrm{best}}/n$, which confines the binary encoding of $\{l_1,\dots,l_j\}$ to two short windows for each δ. For $n \le 2^{20}$ the pruned and the full enumeration agree for every n. The pruned scan of all $n \le 2^{28}$ takes a few minutes on six cores and finds non-uniqueness exactly for the six values of n listed in the introduction, each with two minimizing types.

Remark 5.1. All six ties have the same structure as the pair (T_A, T_B). For a depth δ and a set $L \subseteq \{15,\dots,\delta-2\}$, let $\mathcal{A}_L$ be the type of depth δ with gaps $\{8,\dots,13\}\cup L$ and 13 leaves of depth $\delta-1$, and let $\mathcal{B}_L$ be the type of depth $\delta+1$ with gaps $\{15\}\cup(L+1)$ and 496 leaves of depth $\delta+1$. Kraft's equality shows that both have $n = 2^\delta - 16135 - \sum_{l\in L} 2^l + |L|$ leaves. With respect to the level δ both have $R_\delta = 692 + \sum_{l\in L} l^2$, while $E_\delta(\mathcal{A}_L) = 76 + \sum_{l\in L} l$ and $E_\delta(\mathcal{B}_L) = \sum_{l\in L} l - 482$. So $D(\mathcal{A}_L) = D(\mathcal{B}_L)$ exactly when $\sum_{l\in L} l = 203$. The six counterexamples are the cases where, in addition, these two types are minimizers: $\delta = 27$ with $L = \{15,16,18,\dots,25\}$ (this is (T_A,T_B)), and $\delta = 28$ with five sets L of ten elements.

Neighbouring values. For $n+1 = 67{,}256{,}580$ the minimizer is unique and of type $T_{n+1;A}$, and for $n-1$ it is unique and of the type of T_B. So this value of n does not contradict the third regularity listed in the Discussion of [1], which concerns the passage from n to $n+1$.

6. Formal verification

The statement and its refutation are formalized in Lean 4 (version 4.33.1) with Mathlib. Trees are ordered binary trees, and the definitions follow the paper:

```
inductive BTree where
  | leaf : BTree
  | fork : BTree → BTree → BTree

def leafDepths : BTree → Multiset ℕ
  | .leaf => {0}
  | .fork l r => (leafDepths l + leafDepths r).map Nat.succ

def depthObjective (T : BTree) : ℤ :=
  ((leafCount T) : ℤ) * (depthSquareSum T) - (depthSum T) ^ 2

def MinimizesDepthVariance (n : ℕ) (T : BTree) : Prop :=
  (leafCount T) = n ∧
    ∀ U : BTree, (leafCount U) = n → (depthObjective T) ≤ (depthObjective U)
```

Here `leafCount`, `depthSum` and `depthSquareSum` are the number of leaves, S and Q, computed from `leafDepths`. Ordering the children does not matter: every tree shape has an ordered representative and all quantities are invariant under swapping children. The main theorem is the negation of the conjecture as stated:

```
theorem minimum_variance_depth_unique_false :
    ¬ (∀ (n : ℕ), 1 ≤ n → ∀ (T T' : BTree),
```

```
    MinimizesDepthVariance n T → MinimizesDepthVariance n T' →
    leafDepths T = leafDepths T')
```

The trees T_A and T_B are given as explicit terms built from perfect trees along a path, as in Section 3; their numbers of leaves and their values of S and Q are computed symbolically. Minimality is proved along the lines of Section 4, for an arbitrary `BTree` with n leaves. The cut at level k and the bounds (2)–(4) are proved by induction on trees. The normal form uses the minimum of a feasible set (`Nat.find`) and the replacements of Lemma 4.2. The 16,800 evaluations of (8) are carried out by a Boolean checker whose soundness is proved in Lean, and the checker's run is verified by the kernel (`decide +kernel`). No `sorry`, `native_decide`, additional axioms or metaprogramming are used. The whole file checks in about 45 seconds, and the main theorem depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`.

Author contributions. GPT-6-Astra, acting as an advisor surveying open problems in mathematical biology, proposed the conjecture as a target. It then found the counterexample and the proof of minimality and wrote the Lean formalization and the certificate. Claude Fable 5.1 refereed the proof and the formalization against the definitions of [1], and independently re-checked the certificate, the enumeration of the family of Theorem 2.1, the normal form, and the small and neighbouring cases. Claude Opus 5.5 re-checked the formalization and the computations, carried out the scan for $n \leq 2^{20}$, and wrote the manuscript. A human operator initiated the project and provided computing resources; the mathematics, the formal proofs and the text were produced by the models.

References

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